Solutions
QA : Numbers - 02

1. 2 The given expression can be written as 15N + 13 + 36/N. Thus, the possible values of N for which the given expression is a natural number are 1, 2, 3, 4, 6, 9, 12,18 and 36 [factors of the number 36].

2. 2 $\frac{n^2 7n+12}{n^2-n-12}=\frac{(n+4)(n+3)}{(n-4)(n+3)}=\frac{(n+4)}{(n-4)}$

From the options,

(i) Put n = 16, $\frac{(n+4)}{n-4}=\frac{16+4}{16-4}=\frac{5}{3}$

(ii) Put n = 12, $\frac{n+4}{n-4}=\frac{12+4}{12-4}=2$ → is a positive integer.

∴ The largest positive integer, n = 12.

3. 2 From 12 to 40, there are 7 prime numbers, i.e., 13, 17, 19, 23, 29, 31 and 37 such that (n – 1)! is not divisible by any of them.

4. $N = 2^7 \times 3^6 \times 5^3$

The powers of 2, 3 and 5 are (0, 1, 2, 3, 4, 5, 6, 7), (0, 1, 2, 3, 4, 5, 6) and (0, 1, 2, 3).

a. Total number of factors is 8 × 7 × 4 = 224.

b. For odd factors take only 1 power of 2 i.e. 0 power. Hence, required number of factors = 1 × 7 × 4 = 28.

c. Numberof even factors = total factors – odd factors = 224 – 28 = 196.

d. 3 factors of N are prime i.e. 2, 3, and 5. One factor is neither prime nor composite i.e. 1. So total number of composite number factors is 224 – 4 = 220.

e. a factor will be a multiple of 3 if it contains at least one power of 3. So out of the powers mentioned in N, one power of 3 (i.e. 0 power) is not considered. So number of factors will be 8 × 6 × 4 = 192.

f. $12 = 2^2 \times 3$. To make a number a multiple of 12 we need to take at least 2 powers of 2 and at least 1 power of 3. So power sof 2, 3 and 5 now allowed are (2, 3, 4, 5, 6, 7), (1, 2, 3, 4, 5, 6) and (0, 1, 2, 3). Number of factors divisible by 12 are 6 × 6 × 4 = 144.

g. Perfect squares will have only even powers of the primes. So powers of the primes must be (0, 2, 4, 6), (0, 2, 4, 6) and (0, 2) respectively. So number of factors that are perfect square is 4 × 4 × 2 = 32.

h. Perfect cubes will have only those powers which are divisible by 3 so powers used to make perfect cube factors are (0, 3, 6), (0, 3, 6) and (0, 3) respectively.
So the number of factors that are perfect cube is 3 × 3 × 2 = 18.

i. If the unit digit of a number is 0 it must be a multiple of 2 as well as 5. So power 0 will be omitted from powers of 2 and 5.
Hence, number of factors with unit digit 0 = 7 × 7 × 3 = 147.

j. For last 2 digits to be 0 the number must contain at least 2 powers of 2 and 5 both. Hence such factors will be 6 × 7 × 2 = 84.

k. Sum of the factors is $(2^0 + 2^1 + 2^2 \ldots + 2^7)(3^0 + 3^1 + \ldots + 3^6)(5^0 + 5^1 + 5^2 + 5^3) = 87129588$.

l. Product of the factors = $(N)^{\frac{\text{Number of factors}}{2}} = N^{112}$

5. $N = 2^2 \times 2^4 \times 3^4 \times 5^6 = 2^6 \times 3^4 \times 5^6$

a. Total number of factors of N is (6 + 1) × (4 + 1) × (6 + 1) = 245. If N is written as a product of two numbers, both of them must be factors of N. Here N is a perfect square so one way will be $\sqrt{N} \times \sqrt{N}$. For other 244 factors, there will be 244/2 = 122 pairs. So a total of 123 pairs.

b. Two different ways will be 122 as $\sqrt{N} \times \sqrt{N}$ will not be allowed now.

c. As N is an even number, it cannot be written as a product of 2 odd numbers.

d. Let N = 2a × 2b. Hence $2^4 \times 3^4 \times 5^6 = a \times b$. Again a and b both must be the factors of $2^4 \times 3^4 \times 5^6$ and for each value of a there must be a unique value of b. So total of 5 × 5 × 7 = 175 ways. Out of these 175 ways, there will be exactly 1 way when both a and b are equal. In other 174 ways, the unordered ways will be 174/2 = 87. Hence total number of ways is 88.

e. As one of them is even so all powers of 2 must be there in that number only. Let $N = (2^6 \times a) \times b$. So our answer would be same as the total number of factors of $3^4 \times 5^6$ i.e. (4 + 1) × (6 + 1) = 35.

f. A number N can be written as a product of 2 co-prime numbers in 2^{P-1} ways where P is the number of prime factors. Hence there are 2^{3-1} = 4 ways.

6. 2 Only squares of primes have 3 factors. $720 = 2^4 \times 3^2 \times 5$. It is divisible by two squares of prime i.e. 2^2 and 3^2. Hence, answer is 2.

7. 4 Let N is the number which has 72 factors. Let $N = p_1^a \times p_2^b \times p_3^c$..... where p_1, p_2, p_3 .... Are all primes. So (a + 1)(b + 1)(c + 1) ... = 72. To have minimum primes take a = 71 so only one prime is used. To have maximum primes we need to break 72 into maximum possible factors i.e. 72 = 2·2·2·3·3. So there are 5 prime factors in this case.

8. 4 The prime factoriation of
$2^8 \times 3^6 \times 5^4 \times 10^5$ is $2^{13} \times 3^6 \times 5^9$.
120 can be prime factorised as $2^3 \times 3 \times 5$
All factors of $2^{13} \times 3^6 \times 5^9$ than can be written as multiples of 120 will be of the from $2^3 \times 3 \times 5 \times k$.
∴ $2^{13} \times 3^6 \times 5^9 = 2^3 \times 3 \times 5 \times k$
⇒ $k = 2^{10} \times 3^5 \times 5^8$.
∴ Number of factors of k = (10 + 1) (5 + 1) (8 + 1)
= 11 × 6 × 9 = 594

9. 44 $2^4 \times 3^5 \times 10^4$
$2^4 \times 3^5 \times 2^4 \times 5^4$
$2^8 \times 3^5 \times 5^4$
For the factor to be a perfect square, the factor should be even power of the number.
In 2^8 , the factors which are perfect squares are = 5
Similarly, in 3^5, the factors which are perfect squares are = 3
In 5^4, the factors which are perfect squares are = 3
Number of perfect squares greater than
$1 = 5 \times 3 \times 3 - 1 = 44$.

10. 3 If a number has 4 factors then it would be either of the type p^3 or $p_1 \times p_2$.
Case 1: If the number is p^3 then the factors are 1, p, p^2 and p^3. Acc to the question $1 + p + p^2 = 31$. Only p = 5 satisfies it. The number is 125.
Case 2: If the number is $p_1 \times p_2$ then the factors are 1, p_1, p_2, $p_1 \times p_2$. So $1 + p_1 + p_2 = 31$. This means sum of the two primes p_1 and p_2 is 30. The numbers satisfying this are 7 × 23, 11 × 19 and 13 × 17 i.e. 161, 209 and 221.
Sum of all such numbers is 125 + 161 + 209 + 221 = 716.

11. 3 4t + 1 is an odd number this means powers of 2 cannot be used in the factors. 3 is of the type 4k – 1 and 5 is of the type 4k + 1 hence any power of 5 can be used but only even powers of 3 must be used. So powers of 3 are 0, 2 and 4 only whereas powers of 5 that can be used are 0, 1, 2 and 3. So total number of factors of the type 4t + 1 are 3 × 4 = 12. One of these factors is 1 which is 4 × 0 + 1 but t has to be a natural number so answer 12 – 1 = 11.

12. 4 If there are 9 factors below the square root, there will be 9 above the square root. Here if N is a perfect square, then square root of N will also be a factor otherwise not. So N will have 18 or 19 factors.

13. 91 If there are 9 factors below the square root, there will be 9 above the square root. Here N is a perfect square hence square root of N will also be a factor so N will have 19 factors.
If a number has 19 factors, it must be of the type p^{18} where p is a prime number. This is because 19 cannot be written as a product of 2 numbers. Hence $N^5 = p^{90}$. So the number of factors is 90 + 1 = 91.

14. The number of ways in which a number can be written as a sum of two or more consecutive natural numbers is one less than the number of odd factors of the number. $1000 = 2^3 \times 5^3$.
So number of ways = 3 + 1 – 1 = 3.

15. 4 Let number of elements in progression be n, then
$100 = 1 + (n - 1)d \Rightarrow (n - 1)d = 99 = 3^2 \times 11$
Possible values of d are 1, 3, 9, 11, 33.
Hence, 5 progressions are possible.

16. 5 $0.25 \le 2x \le 200$
So, the possible values of x = –2, –1, 0, 1, 2, 3, 4, 5, 6, 7.
But $2^x + 2$ is perfectly divisible by either 3 or 4, so, on putting x = 0, 1, 2, 4, 6 in $2^x + 2$, the numbers obtained will be perfectly divisible by either 3 or 4. Hence, 5 values of x are possible.

17. 7 Given: $2.25 \le 2 + 2^{n+2} \le 202$
Since, $2 + 2^{n+2} \le 202$, the highest value of 'n' that satisfies the condition is 5.
Because $2^7 = 128$, but $2^8 = 256$.
Therefore, $n + 2 = 7 \Rightarrow n = 5$.
Further, subtracting 2 from the first two parts of the inequality,
$0.25 \le 2^{n+2}$ Or, $2^{-2} \le 2^{n+2}$
Hence, $n \ge -4$
So, from the inequality we can say that the integral value of n can be [–4, 5].
For $3 + 3^{n+1}$ to have integral values, n must be greater than or equal to –1 and go up till 5.

18. 2 Given that: $a \le b$ and $ab = 4^{2017} = 2^{4034}$
$a = 1 = 2^0, b = 2^{4034}$
$a = 2^1, b = 2^{4033}$
$a = 2^2, b = 2^{4032}$
……
……
$a = 2^{2017}, b = 2^{2017}$
Hence, total number of pairs (a, b) of positive integers = 2018.

19. 82 Given A is the largest +ve integer that divides
$N = (3^k + 4^k + 5^k)$
For k = 1; N = 12
k = 2 N = 50
k = 3 N = 216
∴ A = 2
Given that B is the largest +ve integer that divides
$M = 4^k(1 + 3 + 16)$
For k = 1 $M = 20 \times 4$
k = 2 $M = 20 \times 4^2$
k = 3 $M = 20 \times 4^3$
∴ B = 80
Hence, (A + B) = 2 + 80 = 82.

20. 150
Let the number of students in the school be N.
N < 5000
N leaves a remainder of 4 when divided by 9, 10, 12, or 25.
N leaves a remainder of 4 when divided by LCM (9, 10, 12, 25) = 900.
N leaves a remainder of 4 when divided by 900.
N = 900x + 4
Since N < 5000, x can take range from 0 to 5.
But 900x + 4 is a multiple of 11 only when x = 2.
So N = 900 × 2 + 4 = 1804
Since 1804 = 12 × 150 + 4
Hence, when we divide these 1804 students into groups of 12, we get 150 groups.